



# Bikol Dictionary: The Living Archive

## Project Showcase & Architectural Overview



### Vision: Bridging Eras

The Bikol Dictionary is not merely a translation tool; it is a **Living Archive**. It serves as a professional, full-stack platform that bridges the academic precision of 20th-century lexicography with the modern accessibility of AI-driven web and mobile applications.

*"To preserve a language is to preserve a soul. We treat every lemma with the reverence of a 400-year-old manuscript, yet delivered with the speed of light." — The Sovereign Architect*



### Core Pillars of Excellence

#### 1. Academic Rigor (The Mintz Legacy)

At its heart, the project implements a normalized linguistic architecture based on the authoritative works of Malcolm W. Mintz.

- Normalized Data Models:** Strict separation of Roots, Definitions, Conjugations, and Example Sentences.
- Automated Extraction:** Sophisticated Python pipelines parse complex focus types (Actor/Object) and series (Ability/Causative) directly from linguistic texts.
- Linguistic Integrity:** Integration with Wiktionary for Central Bikol lemmas, etymologies, and IPA pronunciations.

#### 2. AI-Enriched Intelligence

We leverage the **Groq API** (Qwen/Qwen3-32b) to breathe life into static data.

- Contextual Generation:** Automated creation of natural Bikol and English example sentences.
- Cross-Lingual Bridge:** Identification of missing Tagalog translations and regional dialects.
- Data Auditing:** AI-driven confidence scoring to ensure the highest integrity before publication.

#### 3. Interactive Functional Fluency

The platform transforms passive lookup into active learning.

- **Verb Conjugator:** Real-time morphological breakdowns for complex Bikol verb structures.
- **Substitution Drills:** Interactive grammar modules based on the Mintz sentence patterns.
- **Spaced Repetition:** Integrated flashcard systems for long-term vocabulary retention.

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## The Tech Stack

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| Layer    | Technologies                                     |
|----------|--------------------------------------------------|
| Frontend | Next.js 16 (App Router), React 19, Framer Motion |
| Styling  | Tailwind CSS v4 (Vanilla CSS Philosophy)         |
| Mobile   | Capacitor (Native Android/iOS Bridge)            |
| Database | Supabase (PostgreSQL), Prisma 7                  |
| Pipeline | Python 3.10+, BeautifulSoup4, aiohttp            |
| Runtime  | Bun (Package Management & Test Runner)           |

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## Design Philosophy: The Living Archive

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The visual identity balances **Resting Rigor** with **Responsive Bloom**.

- **Resting Rigor:** 1px borders and deep zinc neutrals create a structured, authoritative baseline.
- **Responsive Bloom:** Interactive elements "come alive" on hover with **Bicolano Sea Blue** shadows and subtle lifts.
- **Accessibility:** WCAG AA compliance, semantic HTML ( `<dl>` , `<dt>` , `<dd>` ), and touch-target optimization (44x44px).

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## Agentic Governance

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The repository is maintained by a specialized team of AI subagents, ensuring consistency across all domains:

- **Archive Designer:** Enforces visual and UX standards.

- **Data Specialist:** Governs the scraping and enrichment pipelines.
  - **Bikol Expert:** Guards linguistic and cultural integrity.
  - **DB Architect:** Manages schema evolution and integrity.
  - **Mobile Expert:** Optimizes for the native mobile experience.
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## Future Roadmap: "Opus 2026"

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- **Multi-Dialect Expansion:** Deep integration of Albay, Rinconada, and Sorsogon dialects.
  - **Community Contribution Portal:** A dignified, peer-reviewed system for community-sourced lemmas.
  - **Audio Synthesis:** High-fidelity text-to-speech for regional Bikol accents.
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